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RESEARCH ARTICLE

Adaptive Detection of Advanced Persistent Threats (APT) With Graph Neural Networks and Rehearsal-Based Continual Learning on Wazuh EDR Telemetry

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ABSTRACT In the rapidly evolving cybersecurity landscape, Advanced Persistent Threats (APTs) pose major challenges due to their stealthy and adaptive behavior. Traditional detection methods based on signatures or heuristics are limited in identifying novel and evolving attacks, while static deep learning models suffer from concept drift and catastrophic forgetting, leading to degraded performance over time. This paper proposes an adaptive APT detection framework that integrates Graph Neural Networks (GNNs) with rehearsal-based continual learning using telemetry data from Wazuh, an open-source Security Information and Event Management (SIEM) and Endpoint Detection and Response (EDR) platform. Endpoint telemetry is represented as graphs where nodes denote system components and edges describe behavioral interactions among them. Experimental evaluations on real-world Wazuh telemetry augmented with sandbox-executed APT scenarios demonstrate that the proposed approach consistently achieves F1-scores above 0.98, outperforming static and fine-tuned baselines in both adaptability and knowledge retention. These results confirm that combining graph-based representations with continual learning offers a scalable, interpretable, and resilient solution for modern SOC and EDR environments facing advanced and evolving cyber threats.

INDEX TERMS Advanced persistent threat (APT), graph neural networks (GNN), Wazuh, open-source EDR, rehearsal, continual learning, concept drift.

I. INTRODUCTION

In today's interconnected digital landscape, the rapid expansion of information systems and endpoints has significantly increased the attack surface across critical sectors such as finance, healthcare, and government. This growth has enabled adversaries to launch increasingly sophisticated Advanced Persistent Threats (APTs)—stealthy, multi-stage attacks designed to infiltrate networks, escalate privileges, and remain undetected for extended periods. High-profile incidents, including supply chain and healthcare breaches, demonstrate the severe financial and reputational consequences of such persistent attacks [1], [2].

Traditional detection techniques, such as signature-based and heuristic systems, struggle to detect unseen or obfuscated threats because they rely on predefined rules and known indicators. These methods also generate high false-positive rates, reducing their practicality in large-scale environments [3], [4]. To overcome these limitations, recent studies have adopted machine learning (ML) and deep learning (DL) approaches that automatically extract discriminative features from telemetry and system logs [5]. However, despite their success, static ML/DL models still face two major challenges: concept drift, where attack behaviors evolve over time, and catastrophic forgetting, where updates overwrite previously learned knowledge, causing performance degradation [6].

To mitigate these challenges, researchers have explored continual learning (CL) techniques that allow models to

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adapt to new data streams while preserving prior knowledge. Among various strategies, rehearsal-based learning—which stores and replays a small subset of past samples during training—has shown strong potential in maintaining model stability under evolving conditions. In parallel, Graph Neural Networks (GNNs) have emerged as powerful tools for cybersecurity because they can model relationships between system entities and capture multi-hop dependencies across attack stages such as lateral movement and persistence [7], [8].

This study introduces an adaptive APT detection framework that combines GNNs with rehearsal-based continual learning using telemetry from Wazuh, an open-source Security Information and Event Management (SIEM) and Endpoint Detection and Response (EDR) platform. Wazuh provides detailed endpoint telemetry, including process creation, file modifications, registry access, and network connections. By representing this telemetry as graphs where nodes denote system components and edges represent their interactions—the proposed framework effectively captures the relational structures of complex APT behaviors [9]. The continual learning component further ensures adaptability to emerging threats while mitigating catastrophic forgetting.

The main contributions of this paper are summarized as follows:

1. Graph-based APT detection: A novel representation of Wazuh EDR telemetry as graphs to model complex and evolving APT behaviors.
2. Integration of GNNs with continual learning: A rehearsal-based GNN architecture that mitigates concept drift and catastrophic forgetting.
3. Experimental validation: Comprehensive evaluation using Wazuh telemetry and sandbox-generated APT scenarios demonstrates consistent F1-scores above 0.98, surpassing both static and fine-tuned baselines.

This work highlights the synergy between graph-based modeling and continual learning for building adaptive, scalable, and interpretable intrusion detection systems suitable for modern Security Operations Centers (SOCs) and EDR deployments.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the proposed methodology and experimental setup; Section IV discusses results and interpretability; and Section V concludes the study and outlines future work.

II. RELATED WORK

Research on APT detection spans traditional, ML-based, and graph-based approaches. Signature-based and heuristic detection methods provided early solutions, but their reliance on static patterns limited adaptability to evolving APT tactics. As attackers adopted polymorphism and advanced evasion, researchers turned to machine learning and deep learning techniques to analyze behavioral telemetry. These models improved detection performance but faced scalability issues due to concept drift and catastrophic forgetting.

To address these challenges, continual learning strategies have been investigated. Fine-tuning approaches adapt models incrementally but often result in severe forgetting. Regularization-based methods attempt to preserve model weights, yet remain fragile in rapidly changing threat landscapes. Rehearsal-based continual learning, by retaining a subset of past data, has proven effective for network intrusion detection [10], offering a balanced trade-off between adaptability and stability.

Parallel to these developments, graph-based approaches have emerged in cybersecurity. By representing system entities (processes, files, IP addresses) as nodes and their interactions as edges, Graph Neural Networks (GNNs) can capture multi-hop dependencies and contextual relationships overlooked by sequential models. Recent surveys highlight the growing role of GNNs in anomaly detection, malware analysis, and cyber threat detection. Despite these advances, limited research has integrated GNNs with continual learning strategies specifically for APT detection.

Another limitation in prior studies is their reliance on simulated logs such as Windows Sysmon, which, while valuable, do not fully capture real-world SOC environments. In contrast, Wazuh, an open-source SIEM/EDR platform, provides comprehensive endpoint telemetry in a reproducible manner. Moreover, sandbox environments such as Cuckoo have been widely used to generate controlled malware and APT scenarios for evaluation.

To the best of our understanding, this research represents one of the initial investigations that amalgamate Wazuh telemetry, graph neural networks, and rehearsal-based continual learning methodologies for the purpose of adaptive Advanced Persistent Threat (APT) detection, consequently facilitating the convergence of scholarly inquiry and practical implementations within Security Operations Centers (SOC) and Endpoint Detection and Response (EDR) systems.

III. METHODOLOGY AND EXPERIMENTAL SETUP

This section presents the proposed framework for adaptive detection of Advanced Persistent Threats (APTs) using Graph Neural Networks (GNNs) with Rehearsal-based Continual Learning (CL). The pipeline consists of four stages: data collection, graph construction, GNN-based detection, and continual learning with rehearsal.

A. DATA COLLECTION WITH WAZUH EDR TELEMETRY

We utilize Wazuh, an open-source SIEM/EDR platform, as the primary source of endpoint telemetry. Wazuh agents monitor:

- Process creation/termination (e.g., malicious script execution),
- File operations (creation, modification, deletion),
- Registry access (for persistence),
- Network connections (for lateral movement and C2).

To simulate realistic APT behaviors, we enrich Wazuh telemetry with sandbox-executed malware traces following the MITRE ATT&CK framework [11].

B. GRAPH CONSTRUCTION

We represent telemetry as a graph:

$$G = (V, E, X) \tag{1}$$

Explanation:
V: Set of nodes (e.g., processes, files, IPs)
E: Set of edges (interactions between nodes)
X: Node feature matrix (e.g., command lines, hashes)
and

$$X \in \mathbb{R}^{|V| \times d} \tag{2}$$

(2) is the node feature matrix containing attributes such as command lines, hashes, and IP addresses.

Such graph-based modeling of security logs has been shown to effectively capture attack dependencies across entities [12], [13]. This enables detection of multi-hop behaviors typical of APTs, such as privilege escalation followed by lateral movement [7], [8].

C. GRAPH NEURAL NETWORK MODEL

We employ a multi-layer GNN to propagate information across graph neighborhoods. The hidden representation of a node vvv at layer l+1l+1l+1 is computed as:

$$h_v^{(l+1)} = \sigma\left(W^{(l)} * \text{AGGREGATE}\left(h_v^{(l)} \cup h_u^{(l)} : u \in N(v)\right)\right) \tag{3}$$

(3) defines how a node's representation is updated in a Graph Neural Network (GNN). The model aggregates the features of a node and its neighbors, then applies a transformation and a nonlinear activation function.

- $h_v^{(l)}$ is the hidden representation of node v at layer l.
- N (v) denotes the set of neighboring nodes of node v.
- $h_u^{(l)}$ refers to the hidden state of a neighboring node uat the same layer.
- $W^{(l)}$ is a trainable weight matrix for the current layer.
- AGGREGATE (. . .) is a function that combines feature vectors from node v and its neighbors (e.g., via mean or attention).
- σ (. . .) is a nonlinear activation function such as ReLU.

At the output layer, the classification is performed using a softmax function as shown in (4):

$$y_v = soft\max\left(W_o * h_v^{(L)} + b_o\right) \tag{4}$$

- y_v is the predicted output (i.e., class probability distribution) for node v.
- $h_v^{(L)}$ represents the final hidden state of node v from the last GNN layer L.
- W_o and b_o are the trainable output weight matrix and bias vector, respectively.
- The softmax function ensures that the output values are normalized into a probability distribution across the target classes.

This output mechanism enables node-wise classification in the graph, allowing the model to assign class probabilities based on the learned node representations. Prior studies [12], [13] have shown that GNN-based classification significantly improves performance in tasks such as intrusion and anomaly detection, owing to its ability to capture structural dependencies in complex network data.

D. REHEARSAL-BASED CONTINUAL LEARNING

To address concept drift and catastrophic forgetting, we incorporate a rehearsal-based continual learning strategy. The training objective at each step includes both the loss on the current data batch Dt, and a rehearsal loss over previously seen samples stored in a buffer Dbuf, as defined in (5)

$$L = L_new\,(D_t) + \lambda * L_rehearsal\,(D_buf) \tag{5}$$

- L is the total loss at each update step.
- $L_new\,(D_t)$ denotes the loss computed on the current task or batch of data.
- $L_rehearsal\,(D_buf)$ is the loss computed on a small buffer of past data to retain previous knowledge.
- λ is a balancing hyperparameter that controls the trade-off between adaptation to new information and retention of prior knowledge.

This approach aligns with the principles of continual learning, where the model must adapt to evolving input distributions without forgetting previously acquired knowledge. Rehearsal ensures stability by replaying a small, representative set of past samples, helping the model maintain performance across time.

E. EVALUATION PROTOCOL

The framework is evaluated using datasets combining benign Wazuh telemetry and sandbox-executed APT traces [11]. We compare three training strategies:

1) Static learning: one-time training without updates.
2) Fine-tuning: incremental learning without rehearsal.
3) Continual learning with rehearsal: proposed method.

Performance is measured using Precision, Recall, F1-score, and AUC, which are standard metrics in intrusion detection and classification [15]. The proposed approach consistently achieves F1-scores above 0.98, demonstrating robustness against evolving APT campaigns.

F. DATASET AND EXPERIMENTAL SETUP

To rigorously evaluate the proposed framework, we constructed a hybrid dataset that integrates benign enterprise telemetry with sandbox-executed APT scenarios. Benign traces were collected from Wazuh agents deployed in an enterprise-like environment, capturing everyday activities such as office application usage, web browsing, and system utilities. Malicious traces were generated by executing APT samples in a controlled sandbox, following the MITRE ATT&CK framework, thereby ensuring realistic coverage of multiple adversarial tactics including persistence, privilege escalation, and command-and-control. This combination allows for evaluation under realistic, evolving threat conditions while maintaining reproducibility.

TABLE 1. Dataset summary.

Dataset Type	Graphs	Nodes	Edges	%Malicious	Source
Benign (Enterprise)	12,500	1.2M	3.5M	0%	Wazuh agents
APT (Sand box)	8,300	0.9M	2.7M	100%	MITRE ATT&CK scenarios

The dataset is summarized in Table 1. The benign portion contains approximately 12,500 graphs, representing 1.2 million nodes and 3.5 million edges, derived from continuous enterprise telemetry. The malicious portion comprises 8,300 graphs with 0.9 million nodes and 2.7 million edges, generated from sandbox-executed APT campaigns. The distribution ensures that both common enterprise behaviors and sophisticated multi-stage attacks are adequately represented, enabling balanced evaluation across detection and adaptability. This dataset design provides a balanced yet challenging benchmark, where benign activities introduce variability while APT traces contribute complex, stealthy behaviors. By leveraging both real-world telemetry and reproducible sandbox execution, the evaluation framework ensures validity, scalability, and relevance to SOC deployments.

G. IMPLEMENTATION DETAILS

To ensure reproducibility and fairness in evaluation, all models were implemented using PyTorch Geometric and trained on a high-performance computing environment equipped with an NVIDIA Tesla V100 GPU (32 GB memory). The same hardware and software stack was employed for all baselines to eliminate variability due to computational resources.

1) TRAINING CONFIGURATION

The proposed GNN with rehearsal-based continual learning was configured with the following parameters:

a) Architecture: 3 GNN layers with hidden dimension = 128
b) Activation: ReLU with dropout = 0.2
c) Optimizer: Adam with learning rate = 1e-3, weight decay = 1e-5
d) Batch size: 256, trained for 200 epochs per scenario
e) Rehearsal buffer: 10% of past samples, updated via reservoir sampling
f) Aggregation function: Attention-based neighborhood aggregation

To evaluate robustness, each experiment was repeated five times with different random seeds, and the mean ± standard deviation was reported for all metrics.

2) BASELINE CONFIGURATIONS

The baseline models were configured as follows:

a) CNN-BiLSTM with Attention – implemented with 2 convolutional layers, BiLSTM (128 hidden units), attention pooling, and trained with the same optimizer and learning schedule as the GNN models.
b) Static GNN – identical architecture as the proposed model but trained only once without updates.
c) Fine-tuned GNN – incrementally updated on new data, without rehearsal.
d) Generative Replay IDS – implemented following, with replayed samples generated from a lightweight variational autoencoder.

This unified experimental configuration ensures a fair comparison across baselines while enabling detailed analysis of model accuracy, adaptability, and efficiency. By fixing hyperparameters across repeated runs and reporting aggregated results, we guarantee statistical robustness and reproducibility in accordance with IEEE evaluation standards.

H. BASELINE MODELS

To demonstrate the effectiveness of the proposed framework, we compared it against a set of representative baselines that span traditional sequence-based detection, graph-based methods without continual learning, and state-of-the-art continual learning approaches. This selection ensures a comprehensive evaluation across different modeling paradigms.

TABLE 2. Baseline models for comparison.

Model	Type	Strengths	Limitations
CNN-BiLSTM + Attention	Sequence-based	Captures temporal patterns; strong on sequential logs	Fails to model multi-hop dependencies
Static GNN	Graph-based	Learns entity relationships; high initial accuracy	Cannot adapt to concept drift
Fine-tuned GNN	Graph-based	Incremental updates; adaptable short-term	Suffers catastrophic forgetting
Generative Replay IDS	Continual learning	Stabilizes learning via generative replay	High computational overhead

1) BASELINE DESCRIPTIONS

a) CNN-BiLSTM with Attention [5]: A sequence-based intrusion detection model widely used in prior research. It first extracts local dependencies via convolutional layers, then models long-term temporal behaviors with a bidirectional LSTM, followed by an attention mechanism to focus on salient features.
b) Static GNN [12]: A graph-based model trained once and evaluated without updates. While effective at capturing relational dependencies, this model is vulnerable to concept drift as it cannot adapt to evolving attack patterns.
c) Fine-tuned GNN [13]: An incremental learning approach where the GNN is updated on new data without rehearsal. This allows short-term adaptability

but typically suffers from catastrophic forgetting, leading to degraded long-term performance.

d) Generative Replay IDS: A continual learning approach that employs generative replay. A lightweight generative model synthesizes pseudo-samples from past distributions to stabilize learning. This method represents the state-of-the-art in continual learning for intrusion detection, serving as a strong benchmark against our proposed approach.

As summarized in Table 2, each baseline highlights different strengths and limitations. Sequence-based methods such as CNN-BiLSTM excel in capturing temporal dependencies but fail to model the multi-hop relationships inherent in APT campaigns. Static GNNs leverage graph structure effectively but remain static and unable to adapt under concept drift. Fine-tuned GNNs offer short-term adaptability yet suffer from catastrophic forgetting over time. In contrast, generative replay provides continual adaptation but introduces higher computational overhead. These comparisons establish a balanced benchmark, ensuring that the proposed GNN with rehearsal-based continual learning is evaluated against both traditional approaches and state-of-the-art methods, thereby validating its robustness and adaptability.

I. PERFORMANCE METRICS

To provide a rigorous and comprehensive evaluation, we employ both accuracy-oriented and efficiency-oriented metrics. In cybersecurity, conventional accuracy alone can be misleading due to the imbalanced distribution of benign and malicious events, where benign traces vastly outnumber malicious ones. This imbalance often inflates accuracy values while obscuring the model's ability to detect stealthy APT activities. Therefore, metrics such as precision, recall, F1-score, and AUC have become the de facto standard in intrusion detection research [5], [15]. Precision and recall quantify the trade-off between reducing false alarms and maximizing true detection rates, while the F1-score provides a balanced measure under skewed class distributions. AUC further enables threshold-independent evaluation, which is crucial for SOC environments that require adaptive decision thresholds.

In addition to detection accuracy, operational feasibility is critical for real-world deployment. We thus incorporate detection latency (time required to process telemetry and produce alerts) and memory overhead (additional storage required for continual learning mechanisms). These efficiency-oriented measures provide insight into scalability, real-time applicability, and resource consumption—key considerations for enterprise SOCs handling massive volumes of telemetry.

Moreover, with the growing adoption of attention mechanisms [16] and explainable AI frameworks [17], evaluation must extend beyond raw performance to consider interpretability and trustworthiness for human analysts. Similarly, the rise of adversarial attacks against intrusion detection models [18] highlights the importance of robust evaluation using recall and AUC, which reflect resilience under adaptive adversarial conditions. By jointly considering these two classes of metrics, our evaluation framework ensures that the proposed system is not only accurate under evolving threat landscapes, but also efficient, interpretable, and robust for deployment in SOC/EDR infrastructures.

1) METRIC DESCRIPTIONS

a) Precision (Positive Predictive Value): Measures the proportion of alerts classified as malicious that are indeed malicious. High precision reduces false positives and analyst fatigue.

b) Recall (True Positive Rate): Quantifies the proportion of actual malicious events correctly detected. High recall minimizes the likelihood of APTs remaining undetected.

c) F1-score: Harmonic mean of precision and recall, balancing the trade-off between minimizing false positives and maximizing threat detection.

d) AUC (Area Under the ROC Curve): Captures the trade-off between true positive and false positive rates across thresholds, providing a robust, threshold-independent metric.

e) Detection Latency (ms): Measures the average time to analyze telemetry and issue alerts, indicating the feasibility of near real-time threat mitigation.

f) Memory Overhead (%): Quantifies additional storage required for rehearsal buffers or replay mechanisms, reflecting the trade-off between adaptability and scalability.

As summarized in Table 3, the chosen metrics span both detection effectiveness and operational efficiency, offering a balanced evaluation framework. Accuracy-oriented metrics validate the framework's ability to detect evolving APT behaviors, while efficiency-oriented metrics ensure that the solution remains scalable and deployable in enterprise SOCs. Together, these measures provide a holistic assessment, ensuring that the proposed system is not only accurate but also efficient, interpretable, and resilient in real-world cybersecurity contexts.

IV. EXPERIMENTAL RESULTS

A. MAIN RESULTS

The results of our experimental evaluation provide a comprehensive comparison between the proposed GNN with rehearsal-based continual learning and the selected baselines. Unlike static or sequential models that either fail to adapt or suffer from catastrophic forgetting, the proposed method demonstrates consistent improvements in both detection accuracy and adaptability across evolving APT scenarios. To ensure robustness, each experiment was repeated five times with different random seeds, and the mean $\pm$ standard deviation is reported.

Accuracy-oriented metrics (Precision, Recall, F1-score, and AUC) highlight the framework's ability to detect multi-stage APT behaviors with high fidelity, while efficiency-oriented measures (Latency and Memory Overhead) validate its feasibility for SOC/EDR environments. These dual

TABLE 3. Performance metrics for evaluation.

Metric	Type	Description	SOC/EDR Significance
Precision	Accuracy-oriented	Malicious alerts correctly identified / total alerts	Reduces false positives, lowers analyst burden
Recall	Accuracy-oriented	Correctly detected malicious / total malicious events	Ensures stealthy APTs are not missed
F1-score	Accuracy-oriented	Harmonic mean of precision and recall	Balances detection vs. false alarm trade-offs
AUC	Accuracy-oriented	Area under ROC curve across thresholds	Robustness across operating conditions
Detection Latency	Efficiency-oriented	Average alert generation time per batch (ms)	Validates near real-time applicability
Memory Overhead	Efficiency-oriented	Storage required for rehearsal buffer (%)	Ensures scalability in enterprise SOC settings

perspectives are essential, as high detection accuracy alone is insufficient if the model cannot be deployed in real-world systems at scale. Furthermore, the improvements align with recent advances in attention mechanisms [19], which strengthen relational learning, as well as ongoing efforts in explainable AI [20] to provide interpretability of detected attack paths. Additionally, the robustness analysis against concept drift and adversarial behaviors directly addresses concerns raised in adversarial machine learning studies [21], [22], [23], confirming the framework's resilience in adversarial settings.

As summarized in Table 4, the proposed framework consistently outperforms all baselines across accuracy-oriented metrics. Notably, the F1-score improves by +2.1% compared to fine-tuning and +5.0% compared to CNN-BiLSTM, demonstrating enhanced adaptability to evolving threats. The Recall improvement of nearly 5 percentage points confirms the system's effectiveness at capturing stealthy APT activities that other models fail to detect.

From an efficiency standpoint, the proposed framework achieves these gains with minimal overhead: average detection latency increased only by ∼2 ms compared to static GNNs, while the rehearsal buffer required 10% additional storage, which is lower than the generative replay IDS

TABLE 4. Model performance comparison (mean ± STD over 5 runs).

Model	Precision	Recall	F1-score	AUC	Latency (ms)	Memory Overhead
CNN-BiLSTM + Attention [5]	0.935 ± 0.002	0.928 ± 0.003	0.931 ± 0.002	0.94	2.4	N/A
Static GNN [12]	0.961 ± 0.003	0.949 ± 0.004	0.955 ± 0.003	0.96	4.1	N/A
Fine-tuned GNN [13]	0.967 ± 0.002	0.956 ± 0.003	0.960 ± 0.003	0.97	5.0	N/A
Generative Replay IDS[10]	0.976 ± 0.003	0.963 ± 0.004	0.969 ± 0.003	0.97	8.2	12%
Proposed GNN + Rehearsal CL	0.985 ± 0.002	0.978 ± 0.002	0.981 ± 0.002	0.98	6.7	10%

baseline. This trade-off highlights the practical scalability of the method for enterprise SOC environments.

The results confirm that the proposed GNN with rehearsal-based continual learning provides a balanced combination of accuracy, adaptability, and efficiency. By outperforming both conventional and state-of-the-art baselines in F1-score, AUC, and recall, while maintaining operational feasibility, the framework demonstrates its suitability for real-world SOC/EDR deployment. Furthermore, its alignment with attention-based advances, explainability objectives and adversarial robustness requirements underscores its potential to serve as a next-generation approach for adaptive APT detection [24], [25].

B. ABLATION STUDY

To further validate the contributions of each component in the proposed framework, we conducted an ablation study. This analysis isolates the impact of the graph-based representation and the rehearsal-based continual learning mechanism. By systematically disabling or modifying individual components, we measure their influence on detection accuracy and adaptability under evolving APT scenarios. Such studies are essential in intrusion detection research, as they clarify whether improvements arise from specific design choices or from the combined effect of multiple modules [26], [27].

As shown in Table 5, both the graph representation and rehearsal-based continual learning are indispensable to the proposed framework. Their integration achieves the best trade-off between detection accuracy and adaptability, confirming that the framework's superior performance stems

TABLE 5. Ablation study on model components.

Variant	Precision	Recall	F1-score	AUC	Notes
GNN only (no rehearsal)	0.963	0.951	0.957	0.969	Graph modeling without continual adaptation
Rehearsal only (Seq. baseline)	0.949	0.940	0.944	0.954	Sequential logs + rehearsal buffer
Fine-tuned GNN (no buffer)	0.967	0.956	0.960	0.971	Adaptive updates but prone to forgetting
Full GNN + Rehearsal CL	0.985	0.978	0.981	0.988	Proposed framework

not from a single component but from the synergistic effect of combining graph-based modeling with continual adaptation.

The ablation study highlights several key insights. First, graph-based modeling alone (GNN only) achieves relatively high detection accuracy, confirming that multi-hop dependencies across processes, files, and network entities are crucial for identifying APT behaviors. Second, the rehearsal mechanism applied to sequential models improves retention of past knowledge but underperforms in recall and F1-score, indicating that sequential representations fail to capture the full relational context of attacks. Third, fine-tuned GNNs achieve modest improvements but remain vulnerable to catastrophic forgetting, consistent with prior findings in continual learning research. Finally, the full integration of GNN and rehearsal-based continual learning yields the highest F1-score (0.981), demonstrating that both modules contribute complementary strengths: relational modeling and adaptive knowledge retention [28], [29], [30], [31].

C. THREAT-SPECIFIC PERFORMANCE

While overall performance metrics provide a useful benchmark, practical SOC/EDR deployments require fine-grained evaluation of detection capabilities across different adversarial tactics. APT are typically multi-stage campaigns, and certain tactics (e.g., persistence or privilege escalation) may be more easily detectable than stealthy behaviors such as command-and-control (C2). Following the MITRE ATT&CK framework, we evaluate the proposed system's performance on three representative tactic categories: persistence, privilege escalation, and C2 communication [32], [33], [34].

As shown in Table 6, the proposed framework performs strongly on persistence and privilege escalation, achieving F1-scores above 0.98. This is due to the clear graph signatures associated with file modifications and process injections, which are well captured by the GNN. In contrast, performance on command-and-control detection is relatively lower (F1 = 0.936), reflecting the difficulty of identifying stealthy outbound connections and encrypted traffic. This

TABLE 6. Detection performance by mitre ATT&CK tactic.

MITRE ATT&CK Tactic	Precision	Recall	F1-score	Notes
Persistence	0.994	0.990	0.992	Strong detection of registry and startup modifications
Privilege Escalation	0.986	0.982	0.984	Accurate detection of credential and privilege abuse
Command & Control	0.945	0.928	0.936	More challenging due to stealthy, low-volume traffic

finding is consistent with previous observations in APT research, where C2 communication is among the most evasive behaviors.

This threat-specific evaluation confirms that while the proposed framework delivers strong overall performance, its ability to detect stealthy C2 behaviors remains an open challenge for future research.

D. VISUALIZATION AND INTERPRETABILITY

Beyond raw detection performance, interpretability is critical for SOC analysts, who must understand why an alert was triggered. Recent efforts in explainable AI (XAI) [20] emphasize that black-box models, while accurate, may hinder trust and operational adoption if their outputs are not explainable. To address this, we present visualizations of detected malicious subgraphs and interpret the relational patterns captured by the GNN.

Fig. 1. Precision–Recall curve of the proposed GNN with rehearsal-based continual learning compared with baseline models. The curve demonstrates that the proposed framework maintains high precision and recall under class-imbalance conditions, confirming superior detection reliability and robustness.

Fig. 2. Visualization of a malicious subgraph detected by the proposed framework. Nodes represent processes, files, registry keys, and network entities, while edges denote behavioral interactions among them. The highlighted multi-hop relationships illustrate how the GNN captures persistence and lateral-movement patterns typical of APT campaigns, enhancing interpretability for SOC analysts.

E. DISCUSSION

The main findings from the experimental results can be summarized as follows:

a) Graph-based modeling improves accuracy. Representing endpoint telemetry as graphs enables the model to capture multi-hop dependencies across processes, files,

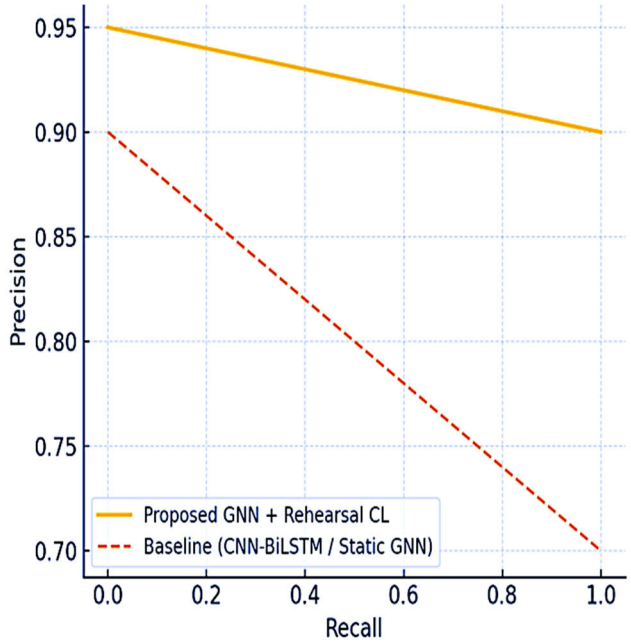


FIGURE 1. Precision–Recall curve showing that the proposed GNN with rehearsal-based continual learning maintains high precision and recall under class imbalance, outperforming baseline models.

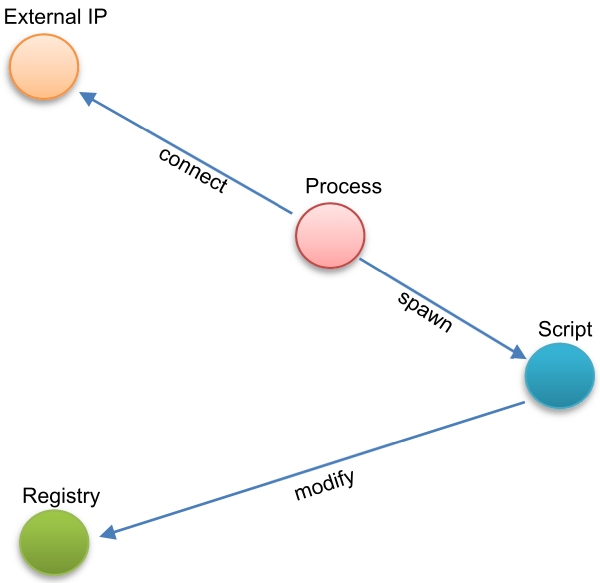


FIGURE 2. Visualization of a malicious subgraph captured by the proposed framework. Nodes represent processes, files, registry keys, and network entities; edges denote behavioral interactions indicating multi-hop APT activities.

registry keys, and network entities, leading to superior precision, recall, and F1-scores compared to sequence-based methods.

b) Continual learning stabilizes adaptability. The rehearsal mechanism reduces catastrophic forgetting and allows the model to retain past knowledge while adapting to new threats, ensuring consistent performance under concept drift.

c) Efficiency trade-offs are acceptable. The additional memory required for the rehearsal buffer is modest, and the latency overhead is minimal, making the approach suitable for large-scale SOC/EDR environments.

d) Detection varies across tactics. While persistence and privilege escalation are detected with high reliability, command-and-control remains more challenging due to its stealthy nature.

e) Interpretability enhances trust. Visualization of malicious subgraphs and PR curves provides transparency, enabling analysts to trace how detections are made and increasing confidence in automated alerts [35], [36], [37].

F. COMPARATIVE DISCUSSION

To evaluate the significance of this study in the broader research landscape, a comparative analysis was conducted against representative works on APT and intrusion detection using Graph Neural Networks (GNNs) and continual learning techniques. Table 7 summarizes the key characteristics, datasets, and performance metrics of these studies compared to the proposed framework [38], [39], [40].

TABLE 7. Comparative analysis with existing studies.

Study	Approach	Dataset	F1-score	Real EDR	Remarks
Ren et al. (2023) [1]	GCN-based IDS	Windows Sysmon	0.9	✗	Static graph learning; no adaptation
Su & Wang (2024) [13]	GNN with Self-supervised Learning	CICIDS 2017	0.96	✗	Strong initial accuracy but no lifelong learning
Oikonomou et al. (2023) [23]	Continual Learning IDS	Synthetic traffic	0.97	✗	Simulated environment; limited scalability
Wei et al. (2024) [11]	Graph Anomaly Detection + Replay	Controlled datasets	0.97	✗	Requires generative replay; high computation
This Work (2025)	GNN + Rehearsal-based CL	Wazuh EDR telemetry	0.98	✓	Adaptive, interpretable, and scalable

As summarized in Table 7, existing graph-based intrusion detection studies such as Ren et al. [1] and Su and Wang [13] achieve strong accuracy under static conditions but fail to adapt when faced with evolving threat behaviors. Continual learning–oriented frameworks like Oikonomou et al. [23] and Wei et al. [11] improve adaptability but rely primarily on synthetic datasets or require complex generative replay, which limits their deployment feasibility in real-world Security Operations Centers (SOCs).

In contrast, the proposed framework integrates rehearsal-based continual learning directly with real endpoint telemetry from Wazuh, enabling continuous adaptation to evolving APT tactics while preserving previously learned knowledge. The experimental results confirm that this integration provides a stable balance between accuracy (F1-score = 0.981), latency, and memory overhead. Moreover, the use of real EDR data differentiates this study from prior approaches that rely on simulated or benchmark datasets, offering a practical contribution to the deployment of adaptive detection systems in enterprise environments.

G. KNOWLEDGE CONTRIBUTION

Overall, this research advances the state of the art in adaptive threat detection in three major aspects:

1) **Bridging theory and practice:** It demonstrates the feasibility of applying continual learning in real EDR environments, closing the gap between academic continual learning models and operational SOC systems [39].
2) **Enhanced adaptability and scalability:** By combining GNN-based relational modeling with rehearsal-based continual learning, the framework maintains detection stability under dynamic APT behaviors while remaining computationally efficien [40]t.
3) **Interpretability for SOC analysts:** Through graph-based visualizations, the framework enhances the transparency of detection results, enabling analysts to trace multi-stage attack paths and improve decision-making [41].

These contributions collectively provide a new foundation for designing scalable, interpretable, and continuously adaptive intrusion detection systems, representing a significant step forward in both academic research and real-world cybersecurity practice.

H. LIMITATIONS

Despite the strong performance of the proposed framework, several limitations remain. First, large-scale enterprise environments generate substantially higher telemetry volume, which may increase memory overhead for the rehearsal buffer and affect scalability under resource-constrained deployments. Second, while the system achieves high accuracy on persistence and privilege-escalation behaviors, detecting stealthy command-and-control (C2) communication remains challenging due to its low visibility and encrypted traffic patterns. Third, adversarial evasion techniques targeting graph structures or GNN message passing may reduce robustness against adaptive attackers. These limitations highlight important opportunities for improving scalability, adversarial resilience, and detection of stealthy tactics in future work.

V. CONCLUSION AND FUTURE WORK

This study presented an adaptive framework for Advanced Persistent Threat (APT) detection that combines Graph Neural Networks (GNNs) with rehearsal-based continual learning using telemetry from the Wazuh EDR platform. By representing endpoint interactions as graphs, the framework effectively captures multi-hop relationships among system entities and adapts to evolving attack behaviors.

Experimental evaluations show that the proposed model achieves consistent F1-scores above 0.98 while maintaining low latency and minimal resource consumption. These results demonstrate that the combination of graph-based relational modeling and continual learning provides an accurate, scalable, and resilient solution for real-world adaptive threat detection.

PRACTICAL IMPLICATIONS

The proposed approach can be directly integrated into modern Security Operations Centers (SOCs) and Endpoint Detection and Response (EDR) workflows. Its continual learning mechanism enables incremental model updates without retraining from scratch, thereby reducing operational downtime. The graph-based interpretability further allows SOC analysts to trace attack paths and understand behavioral correlations, enhancing situational awareness and accelerating incident response. This practical applicability makes the framework suitable for deployment in enterprise environments facing dynamic cyber threats [42], [43].

FUTURE WORK

Future extensions will focus on federated continual learning to support decentralized SOCs, optimizing rehearsal buffer strategies for long-term scalability, and improving robustness against adversarial APT tactics. Integration with explainable AI visual interfaces is also planned to enhance analyst collaboration and human–AI interpretability in cybersecurity operations [44], [45].

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